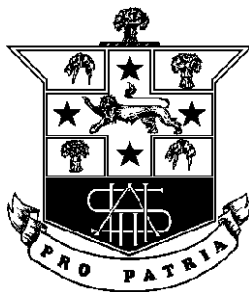


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS – EXTENSION TWO

2009 HSC

ASSESSMENT TASK 1

Examiners ~ G Huxley, G Rawson

GENERAL INSTRUCTIONS

- | | |
|---|---|
| <ul style="list-style-type: none">• Reading Time – 3 minutes.• Working Time – 40 MINUTES.• Attempt all questions.• All necessary working should be shown in every question.• This paper contains two (2) questions. | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved calculators may be used.• Each question is to be started on a new piece of paper.• This examination paper must NOT be removed from the examination room. |
|---|---|

STUDENT NAME: _____

TEACHER: _____

QUESTION ONE **18 marks** *Start a SEPARATE sheet*

- (a) Let $z = 1 + 2i$ and $\omega = 1 + i$. Find, in the form $x + iy$,
- (i) $z \omega$ 1
- (ii) $z \bar{\omega}$ 1
- (iii) $\frac{1}{\omega}$ 1
- (b) z has modulus r and argument θ . Find in terms of r and θ the modulus and argument of:
- (i) z^2 1
- (ii) $\frac{1}{z}$ 1
- (iii) iz 1
- (c) Find all pairs of integers x and y that satisfy $(x + iy)^2 = 24 + 10i$ 3
- (d) Consider the equation $z^2 + az + (1 + i) = 0$
Find the value of a , given that i is a root of the equation. 2
- (e) It is given that two of the roots of $P(z) = z^3 + pz^2 + qz + 32$ are -2 and $4i$.
If it is known that p and q are real numbers, state the third root of $P(z)$,
giving a reason for your answer 1
- (f) Let $z = -2 + 3i$
- (i) Evaluate $\bar{z}$. Verify that $z \bar{z}$ is real. 2
- (ii) Use $\frac{1}{z} = \frac{\bar{z}}{z\bar{z}}$ to find $\frac{1}{z}$ in the form $a + bi$, where a and b are real. 2
- (g) Simplify
$$\frac{\left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12}\right) \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)}{\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)}$$
 2

QUESTION TWO **18 marks** *Start a SEPARATE sheet*

- (a) By considering the complex number $z = x + iy$ in the Argand plane, sketch the locus
Of the following on separate Argand diagrams.

(i) $\arg z = \frac{\pi}{3}$ **1**

(ii) $\arg \bar{z} = \frac{\pi}{3}$ **1**

(iii) $\arg(-z) = \frac{\pi}{3}$ **1**

- (b) Let $w = r(\cos \phi + i \sin \phi)$ where ϕ is an acute angle.
With the aid of a suitable diagram, or otherwise:

(i) Show that the distance between w and $\bar{w}$ in the complex plane is $2r \sin \phi$ **2**

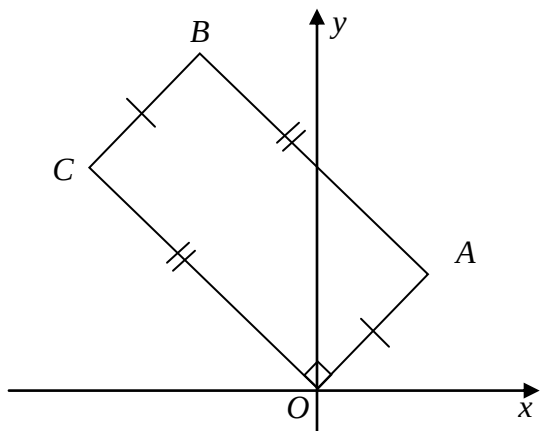
(ii) Find $|w + \bar{w}|$ in terms of r and ϕ . **1**

- (c) Sketch the region of the complex plane for which the complex number $z = x + iy$
has a positive real part and $|z + 3i| \leq 2$. **2**

- (d) Sketch the region defined by $1 < |z - (1 + i\sqrt{3})| < 2$ and $0 \leq \arg z \leq \frac{\pi}{3}$ **3**

Question 2 continued on next page...

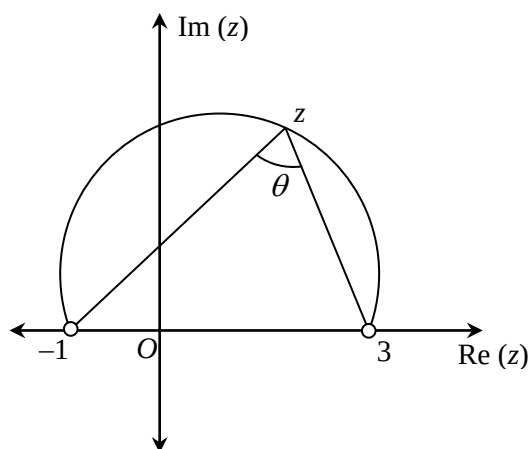
- (e) In the Argand diagram, $OABC$ is a rectangle, where $OC = 2OA$. The vertex A corresponds to the complex number ω .



- (i) What complex number corresponds to the vertex C ? 1
- (ii) What complex number corresponds to the point of intersection D of the diagonals OB and AC ? 2

- (f) The diagram below shows the locus of points z in the complex plane such that $\arg(z - 3) - \arg(z + 1) = \frac{\pi}{3}$.

This locus is part of a circle. The angle between the lines from -1 to z and from 3 to z is θ , as shown.



Copy this diagram into your Writing Booklet

- (i) Explain why $\theta = \frac{\pi}{3}$ 2
- (ii) Find the centre of the circle. 2

Year 12		Mathematics Extension 2	HSC Task 1 Dec 2008
Question No. 1 Solutions and Marking Guidelines			
Outcomes Addressed in this Question			
E3 - uses the relationship between algebraic and geometric representations of complex numbers and of conic sections			
Outcome	Solutions	Marking Guidelines	
E3	(a) (i) $z\omega = (1+2i)(1+i)$ $= -1+3i$	1 mark: Correct answer	
	(ii) $\overline{z\omega} = (1+2i)(1-i)$ $= 3+i$	1 mark: Correct answer	
	(iii) $\frac{1}{\omega} = \frac{1}{1+i} \times \frac{1-i}{1-i}$ $= \frac{1-i}{2}$	1 mark: Correct answer	
	(b) (i) $\text{mod}(z^2) = (\text{mod } z)^2 = r^2$ $\arg(z^2) = 2 \arg(z) = 2\theta$	1 mark: Both mod and arg correct.	
	(ii) $\text{mod}\left(\frac{1}{z}\right) = \frac{1}{\text{mod}(z)} = \frac{1}{r}$ $\arg\left(\frac{1}{z}\right) = -\arg(z) = -\theta$	1 mark: Both mod and arg correct.	
	(iii) $\text{mod}(iz) = \text{mod}(i) \text{mod}(z) = r$ $\arg(iz) = \arg(i) + \arg(z) = \frac{\pi}{2} + \theta$	1 mark: Both mod and arg correct.	
	(c) If $(x+iy)^2 = 24+10i$ Then: $x^2 - y^2 = 24$(1) $2xy = 10$(2) Solving simultaneously for real values of x and y gives: $(x,y) = (5,1) \text{ or } (-5,-1)$ i.e: $x+iy = \pm(5+i)$	3 marks: Correct solution. 2 marks: Solution with one error when calculating the pairs required. 1 mark: Correct pair of simultaneous equations.	
	(d) $P(i) = i^2 + ai + (1+i) = 0$ $i(a+1) = 0$ $\therefore a = -1$	2 marks: Correct solution. 1 mark: Substitution of i followed by some correct simplification.	

(e)	$3^{\text{rd}} \text{ root} = 4i$ Complex roots are in conjugate pairs, since all coefficients are real.	1 mark: Need to name <u>all three</u> components of the solution shown.
(f)	$z \bar{z} = (-2 + 3i)(-2 - 3i)$ (i) $= (-2)^2 - (3i)^2$ $= 13$ which is real.	2 marks: Correct solution. 1 mark: Correct conjugate and significant progress towards solution.
	(ii) $\frac{\bar{z}}{z \bar{z}} = \frac{-2 - 3i}{13}$ $\therefore a = \frac{-2}{13}, b = \frac{-3}{13}$	2 marks: Correct values for a and b . (Correct from answer to (i)) 1 mark: 1 of the 2 values correct.
(g)	$\frac{\left(\text{cis } \frac{5\pi}{12} \right) \left(\text{cis } \frac{3\pi}{12} \right)}{\left(\text{cis } \frac{2\pi}{3} \right)} = \text{cis} \left(\frac{5\pi}{12} + \frac{3\pi}{4} - \frac{2\pi}{3} \right)$ $= \text{cis} \left(\frac{\pi}{2} \right)$ $= i$	2 marks: Correct answer, fully simplified. 1 mark: Correct application of sum and difference of arguments. or: correct simplification after 1 error in initial calculation.